AWS CLOUD COMPUTING

Introduction to AWS • Cloud Services • EC2 Instance Launch

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1. Introduction to Cloud Computing

Cloud computing delivers IT resources over the internet.
Access servers, storage, databases, networking, and software on demand.
Benefits: scalability, flexibility, reliability, and reduced upfront cost.
Service models: IaaS, PaaS, and SaaS.
Deployment models: public cloud, private cloud, and hybrid cloud.

2. Introduction to Amazon Web Services

AWS is a cloud platform offering a wide range of on-demand services.
EC2: virtual servers; S3: object storage; RDS: managed databases.
IAM: identity and access management; VPC: virtual networking.
CloudWatch helps monitor metrics, logs, and alarms.
AWS Regions and Availability Zones support geographic deployment and resilience.

3. AWS Console, Account & IAM

The AWS Management Console is a web interface for managing services.
Choose the correct AWS Region before creating resources.
IAM includes users, groups, roles, and policies.
Apply least privilege: grant only the permissions required.
Protect credentials and enable multi-factor authentication (MFA).

4. Amazon EC2 Overview

Amazon Elastic Compute Cloud (EC2) provides resizable virtual servers called instances.

AMI

Template containing an operating system and launch configuration.

Instance type

Defines CPU, memory, and networking capacity.

Key pair

Supports secure authentication for compatible access methods.

Security group

Virtual firewall that controls network traffic.

EBS volume

Persistent block storage attached to an instance.

IP address

Enables network communication when configured.

5. Steps to Launch an EC2 Instance

01 Open EC2

Sign in to AWS Console and open EC2.

02 Name & AMI

Enter a name and select an operating system image.

03 Instance type

Choose a type suitable for the workload and budget.

04 Key pair

Create or select a key pair where needed.

05 Network & storage

Review VPC, subnet, security group, and EBS.

06 Launch & verify

Launch and confirm the instance is running and checks pass.

6. EC2 Security & Remote Access

Security groups control inbound and outbound traffic.
Common ports: 22 (SSH), 80 (HTTP), 443 (HTTPS), 3389 (RDP).
Restrict SSH/RDP access to trusted IP addresses.
Use supported connection methods such as EC2 Instance Connect or SSH.
Never share private keys or expose credentials in public repositories.

7. Linux Basics & Storage

Useful Linux Commands

pwd — current directory
ls — list files
mkdir — create a directory
cd — change directory
df -h — view disk space
whoami — current user

AWS Storage Services

Amazon EBS: block storage for EC2.

EBS snapshots: point-in-time backups.

Amazon S3: object storage for files, datasets, and backups.

Keep storage access permissions secure.

8. AWS Networking: VPC

VPC: logically isolated network for AWS resources.
Subnet: a range of IP addresses within a VPC.
Route table: directs network traffic.
Internet gateway: enables internet routing when configured.
Security groups and network ACLs help control network access.
A public IP alone does not guarantee connectivity; routes and rules must also be correct.

9. Monitoring, Billing & Cost Control

CloudWatch can monitor metrics, logs, and alarms.
Review AWS Billing and Cost Management regularly.
Use AWS Budgets to configure cost thresholds and alerts.
Stop unused instances when appropriate; terminate resources only when no longer needed.
EBS storage, snapshots, public IPv4 addresses, and data transfer may still incur charges.
Follow least privilege and remove unused resources carefully.

Conclusion & Learning Outcomes

During this internship, I learned:

Cloud computing concepts and key AWS services.
How to configure and launch an EC2 instance.
Fundamentals of IAM, security groups, and VPC networking.
Basics of Linux, EBS, S3, monitoring, and cost management.

Thank You